



The AI that understands your enterprise data

A single platform that connects any data source, builds a live enterprise Knowledge Graph, answers questions in plain language, and runs autonomous agents — with full AI auditability.

Semantic Layer

Knowledge Graph

Natural Language Query

Autonomous Agents

EU AI Act Ready

Manufacturing · Retail · Healthcare · Finance

THE PROBLEM

Every company has the data. Not everyone can use it.

IT answers operational questions

Data engineers write ad-hoc SQL on request instead of building infrastructure that scales.

Decisions based on stale exports

The "single source of truth" is a manually updated Excel sheet. There is no real-time, trustworthy picture.

Data silos everywhere

ERP, CRM, HR, PIM speak different languages. Nobody truly connects them. Every cross-source join is a project.

Weeks waiting for a report

Business teams wait days or weeks for a query that spans two systems. By the time it arrives, the decision is stale.

Compliance without traceability

GDPR, EU AI Act. and

[+ OUR ANSWER](#)

A semantic layer that **unifies, understands, and acts**

Not another BI tool. Not another pipeline. A platform that brings semantic intelligence on top of the data you already have.

Connect any source

SAP, Oracle, Salesforce, PostgreSQL, REST API, CSV. 12+ native connectors. Automatic schema detection. Zero custom ETL.

Enterprise Knowledge Graph

OWL/RDF ontology that models entities, relationships, and cross-system identity resolution rules. Versioned and queryable.

Natural language queries

Neuro-symbolic: deterministic intent detection + Claude AI fallback. SQL always templated, never arbitrary. Zero hallucinations.

Autonomous agents

Monitor, Alert, Reconciler, Validator. Running on the semantic layer with human approval and full audit before any action.

Governance & EU AI Act

Immutable audit trail, granular RBAC, EU data sovereignty, right-to-explanation on every AI output. Compliance by design.

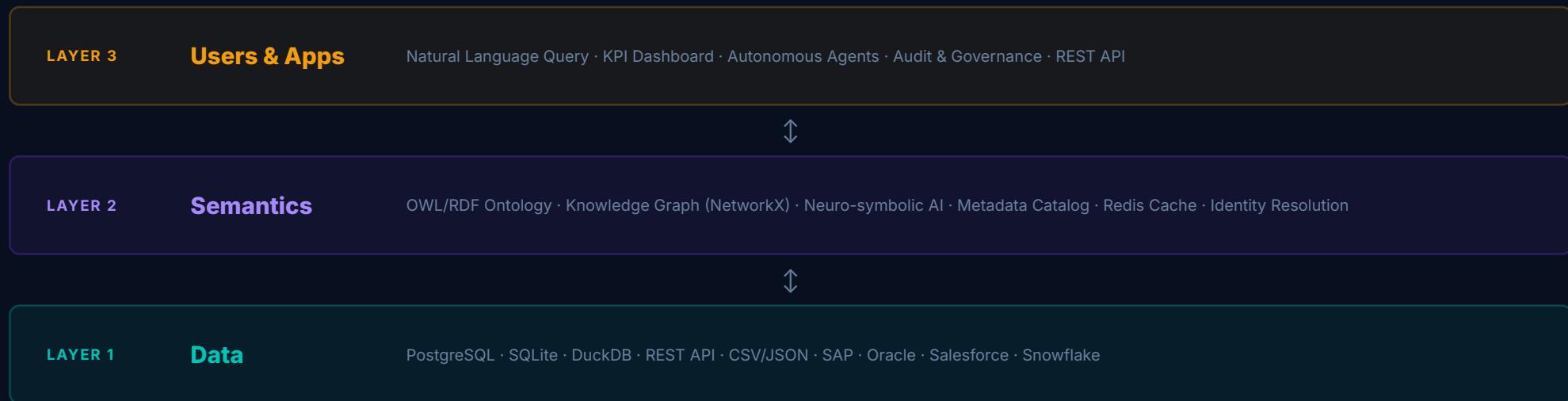
Multi-sector, multi-tenant

Same platform for Manufacturing, Retail, Healthcare, Finance. Domain-specific ontology per customer. API-first, zero lock-in.

ARCHITECTURE

3 layers. One semantic stack.

Every layer is independent, composable, and integrates with your existing stack.



FastAPI + Python 3.11

React 18 + TypeScript

rdflib OWL

NetworkX KG

DuckDB

Redis Cache

JWT Auth HS256

Anthropic Claude

SlowAPI Rate Limiting

FOR IT

Semantic Layer & Knowledge Graph

Give IT a real tool — not another pipeline to maintain by hand.

- Build the enterprise ontology visually: entities, relationships, and typed data properties (XSD)
- Define **identity resolution** rules across systems — e.g. ERP.customer_ref ↔ CRM.accountId
- The Knowledge Graph builds automatically from the ETL pipeline (Connect → Extract → Map → Enrich → Index)
- Versioned semantic mappings: synonyms, aliases, certified metrics (revenue, margin, conversion rate)
- Every mapping change invalidates the Redis cache and updates **zero semantic drift** lineage —
- Neuro-symbolic intent detection: deterministic rules + Claude fallback for ambiguous queries
- SQL is always **templated and signed off** — no arbitrary LLM-generated queries

MANUFACTURING DEMO · ADVENTUREWORKS

193K

KG Nodes

313K

KG Edges

8

Ontology Entities

47

Semantic Definitions

3

Cross-source Bridges

12+

Native Connectors

Modeled Entities

Customer · SalesOrder · OrderLineItem · Product · Employee · Quote · QuoteLineItem · Territory

No black boxes. Every decision traced.

Immutable Audit Trail

Who asked what, which SQL was executed, from which sources, with what latency. Every record locked, every change logged — always.



EU AI Act Ready

Right-to-explanation on every AI output. Configurable EU data sovereignty. Human-in-the-loop mandatory for actions below confidence threshold (<80%).



Granular RBAC

User / admin / manager roles with distinct permissions. Every high-risk write-back action requires explicit manager approval before execution.



Security by Design

Rate limiting (60 queries/min), SQL injection blocking, keyword filtering (DROP/ALTER/DELETE), system table access prevention. JWT HS256.



End-to-End Data Lineage

Every answer includes: tables touched, connectors used, semantic bridges traversed, latency. Traceable all the way back to the raw source.

Certified Enterprise Connectors

SAP S/4HANA · Oracle ERP · Salesforce · Epic EHR (HL7 FHIR) · Snowflake · Anthropic Claude · PostgreSQL · REST/CSV

FOR BUSINESS

Ask. Get answers. Decide.

Business users query their data like talking to a senior analyst — no SQL, no waiting, no tickets.

USER "Who is the top salesperson in Q4 by net revenue?"

SI **Linda Mitchell** — \$4,251,368 YTD · +12% vs Q3 · Territory: Northwest

Automatic JOIN: ERP.salesperson × ERP.sales_order_header × HR.employees via bridge salesperson_ref ↔ EmployeeId

⌚ 145ms

ERP salesperson

ERP sales_order_header

HR employees

bridge: salesperson_ref ↔ EmployeeId

✓ intent: ERP_SALES_TOP

SQL available

Ambiguity handled

If "revenue" maps to two metrics, the system surfaces both candidates and documents the resolution in the semantic layer.

Automatic cross-source joins

Joins across ERP, CRM, HR follow the Knowledge Graph — no SQL written by hand, ever.

Provenance always included

Every answer ships with SQL, sources, latency, lineage, and semantic notes. No black box.

Doesn't just answer. **Acts.** With oversight.

Monitor Agent

Continuously scans KPIs, thresholds, and anomalies across enterprise data. Triggers on schedule or on event.

▶ Sales Performance Monitor active · YTD report delivered

Alert Agent

Notifies teams when data deviates from semantic parameters defined in the ontology.

▶ 3 reps below threshold · alert dispatched to manager

Reconciler Agent

Deduplicates CRM accounts, validates ERP↔CRM bridges, heals inconsistencies across sources.

▶ 372 duplicate accounts removed · 18,484/19,829 customers matched

Validator Agent

Checks referential integrity, semantic constraints, cardinalities, and entity ownership across the ontology.

▶ ERP↔CRM bridge validated · accuracy 99.7%

PROPOSAL



VALIDATION
semantic layer



APPROVAL
manager



EXECUTION



AUDIT LOG
immutable

LIVE DEMO

Manufacturing with **AdventureWorks** — ready today

01

Connect
3 sources

02

Ontology
8 entities

03

Knowledge
Graph

04

Semantic
Layer

05

NL
Query

06

Autonomous
Agents

152K

ERP rows

59K

CRM rows

8.8s

Full pipeline

145ms

Avg NL query

- ✓ [CONNECT] PostgreSQL ERP — connected in **18ms**
- ✓ [EXTRACT] ERP · **152,825** rows extracted · SQLite loaded in **42ms**
- ✓ [MAP] Identity resolution: **18,484 / 19,829** customers matched across ERP ↔ CRM
- ✓ [ENRICH] **372** duplicate CRM accounts removed · **2** semantic ambiguities documented
- ✓ [INDEX] DuckDB materialized · KG: **193,062** nodes · **313,218** edges in **2.1s**

VERTICALS

One platform. **Four sectors.** Same semantic engine.

Manufacturing

ERP

CRM

HR

PIM

MES

- ▶ Cross-source sales performance (ERP × HR)
- ▶ CRM deduplication & bridge validation
- ▶ Supply chain visibility & anomaly detection
- ▶ [Live demo available today with AdventureWorks](#)

Retail

POS

eCommerce

Inventory

CRM

Workforce

- ▶ Unified Customer 360 across all touchpoints
- ▶ Inventory anomaly & stockout prediction
- ▶ Cross-channel promotion effectiveness
- ▶ Workforce scheduling optimized to demand

Healthcare

EHR Epic

HL7 FHIR

Claims

Lab

Scheduling

- ▶ Patient pathway tracking across systems
- ▶ Clinical KPIs & care pathway compliance
- ▶ Claims reconciliation & billing anomaly detection
- ▶ Automated regulatory reporting

Finance

Core Banking

Risk

AML

KYC

Loan

- ▶ Fraud detection across KYC × transactional data
- ▶ AML pattern recognition with AI audit trail
- ▶ Regulatory reporting (EBA, Basel IV)
- ▶ Loan performance & cross-portfolio risk

DIFFERENTIATORS

Why Semantic Intelligence

CAPABILITY	SEMANTIC INTELLIGENCE	TRADITIONAL BI	GENERIC GPT
Natural language query	✓ Neuro-symbolic native	✗ Requires SQL	✗ Generates arbitrary SQL
Automatic cross-source joins	✓ Via Knowledge Graph	✗ Manual, expensive	✗ No ontology
Full AI audit trail	✓ EU AI Act ready	✗ Not applicable	✗ No traceability
Autonomous agents on data	✓ Monitor/Alert/Reconciler	✗ Static dashboards only	✗ No governance
Deterministic SQL (no hallucination)	✓ Signed templates	✓ Direct SQL	✗ Unverifiable output
Zero vendor lock-in	✓ Any DB, API-first	✗ Often proprietary	✗ Depends on API

1 day

Live demo on your use cases

2 weeks

POC on real customer data

60 days

In production with active agents

NEXT STEPS

Three ways to start right now

LIVE DEMO

See it in action

⌚ 45 minutes

Personalized session for your sector. We show AdventureWorks or build a scenario around your specific business questions.

- ✓ Manufacturing / Retail / Healthcare / Finance
- ✓ Real questions from your business
- ✓ Full technical architecture walkthrough

POC

Try it on your data

⌚ 2 weeks

We connect one real source, build the domain ontology, and run 10 business queries. Measurable, documented results.

- ✓ 1 customer data source (DB or API)
- ✓ Domain ontology from scratch
- ✓ Value report with full lineage

PILOT

Value in production

⌚ 60 days

One key domain fully live: semantic layer, active agents, governance configured, and KPIs measured and reported.

- ✓ Autonomous agents configured
- ✓ RBAC & live audit trail
- ✓ Documented ROI at pilot close

What you need for the POC: access to one data source · 2–3 business questions · an IT contact

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